Activity 6

OBJECTIVE

To verify the identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ using a set of unit cubes

Materials Required

A set of 128 plastic cubes each having dimensions (1 unit × 1 unit × 1 unit).

Procedure We shall verify the identity for $a = 3$ and $b = 1$.

Step 1. We shall make **Arrangement 1** for 64 cubes.

This arrangement consists of a stack formed by 16 columns of 4 cubes each, as shown in Figure 6.1. The total volume of this arrangement of cubes is calculated (see the calculations).

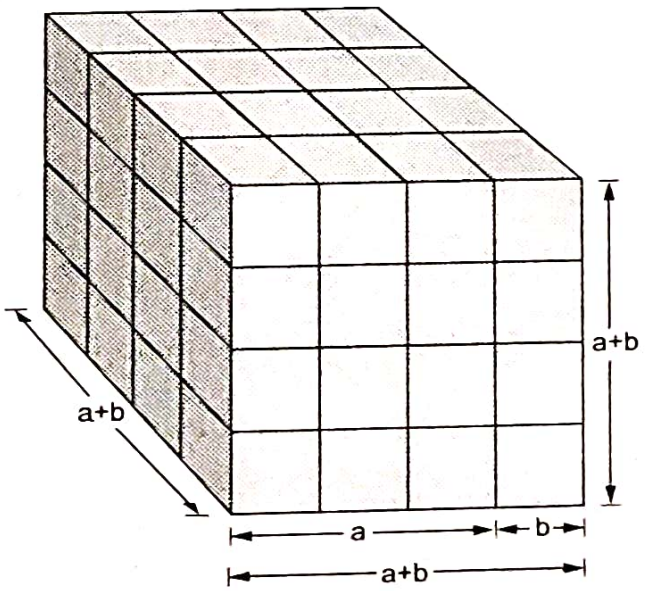

Fig. 6.1

Step 2. We shall make **Arrangement 2** for 64 cubes.

This arrangement consists of 8 stacks. Stack I shown in Figure 6.2(a) consists of 9 columns of 3

cubes each. Stacks II, III and IV, shown in Figure 6.2(b), consist of 3 columns of 3 cubes each. Stacks V, VI and VII, shown in Figure 6.2(c), consist of 1 column of 3 cubes each. Stack VIII shown in Figure 6.2(d) consists of only one cube. The volume of each stack is calculated and these volumes are added up to get the total volume of this arrangement of cubes (see the calculations).

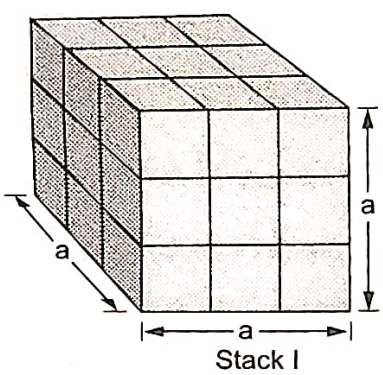

Fig. 6.2(a)

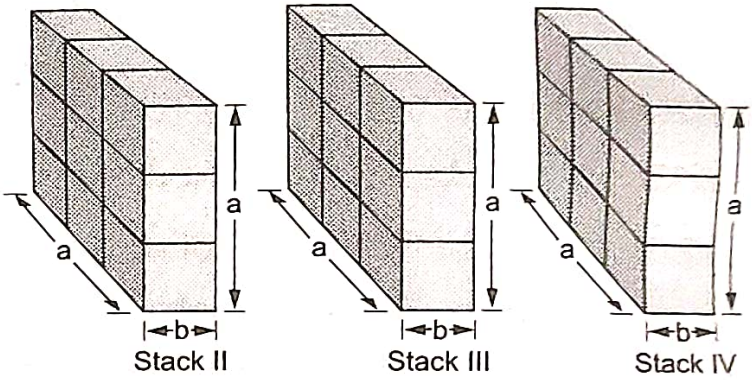

Fig. 6.2(b)

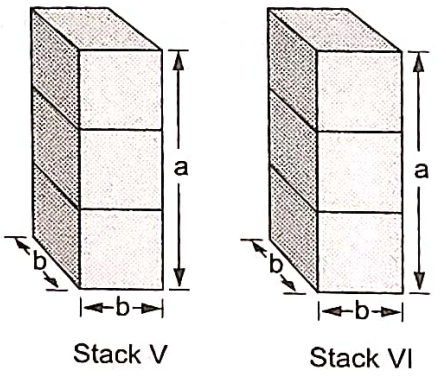

Fig. 6.2(c)

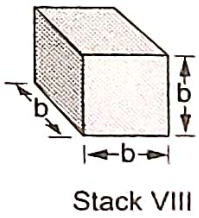

Fig. 6.2(d)

Observations

Since the two arrangements have equal number of cubes (each arrangement has 64 cubes) and all the cubes have the same volume (1 cubic unit), the total volume in both the arrangements must be the same.

Calculations

1. Volume of Arrangement 1

Volume of the stack in Figure 6.1 $=(a+b)^3$.

$\therefore$ volume of Arrangement 1 of the cubes $=(a+b)^3$.

2. Volume of Arrangement 2

Volume of Stack I [stack in Figure 6.2(a)] $=a^3$.

Volume of each of Stacks II, III and IV $=a^2b$.

$\therefore$ total volume of the three stacks [shown in Figure 6.2(b)] $= 3a^2b$.

Volume of each of Stacks V, VI and VII $=ab^2$.

$\therefore$ total volume of the three stacks [shown in Figure 6.2(c)] $= 3ab^2$.

Volume of Stack VIII $=b^3$.

Thus, total volume of Arrangement 2 of the cubes

$$=a^3+3a^2b+3ab^2+b^3.$$

Since the total volume in the two arrangements must be the same, therefore

$$(a+b)^3=a^3+3a^2b+3ab^2+b^3.$$

Result It is verified that $(a+b)^3=a^3+3a^2b+3ab^2+b^3$.

Remarks The students must try to verify this identity for other values of a and b by taking required number of cubes and arranging them suitably.

Activity 22

OBJECTIVE

To verify the midpoint theorem

Materials Required

(a) A piece of cardboard

(b) Two sheets of white paper

(c) A geometry box

(d) A black marker pen

Theory **Midpoint theorem:** The line segment joining the midpoints of any two sides of a triangle is parallel to the third side.

Procedure

Step 1. Paste one sheet of white paper on the cardboard. Draw a ΔABC on this paper.

Step 2. Mark the midpoints D and E of the sides AB and AC respectively (the midpoints of the sides can be obtained by the method of paper folding). Join D and E. Blacken ΔADE with the marker pen.

Step 3. Cut another triangle CEF from the other sheet of white paper so that ΔCEF is congruent to ΔADE. Blacken ΔCEF with the marker pen. Place ΔCEF on the previous paper as shown in Figure 22.1.

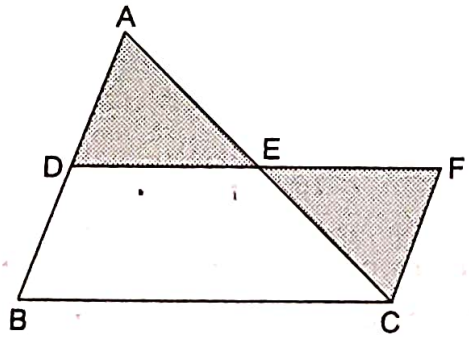

Fig. 22.1

Observations

1. (i) Since ΔCEF is congruent to ΔADE, therefore $DE = EF$.

   (ii) Measure DE and BC. We find that $DE = \frac{1}{2} BC$.

   (iii) From (i) and (ii), we derive that $DF = BC$.

2. (i) Since ΔCEF is congruent to ΔADE, therefore $AD = FC$.

   (ii) Since D is the midpoint of AB, we have $AD = DB$.

   (iii) From (i) and (ii), we get $FC = DB$.

3. From the above observations, it is clear that $DFCB$ is a parallelogram.

   Hence, $DE \parallel BC$.

Result The midpoint theorem is verified.

Activity 33

OBJECTIVE

To derive a formula for finding the area of a circle (Method 2)

Materials Required

(a) Two sheets of white paper

(b) A geometry box

(c) A pair of scissors

(d) A tube of glue

Theory The geometrical formula to determine the area (A) of a circle of radius r is given by $A = \pi r^2$.

Procedure **Step 1.** Draw any circle on a sheet of white paper. Let its radius be r units.

Step 2. Draw eight diameters dividing the circle into 16 equal parts. Shade the alternate parts as shown in Figure 33.1.

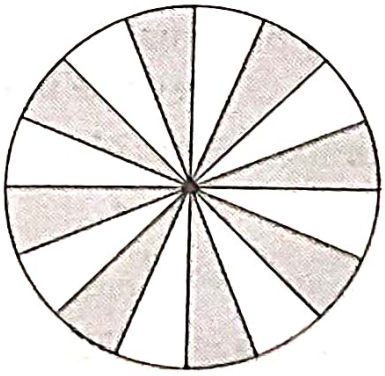

Fig. 33.1

Step 3. Cut these 16 parts of the circle and place them on the other sheet of white paper as shown in Figure

33.2. Here, AB = the circumference of the circle (Figure 33.1) $= 2\pi r$.

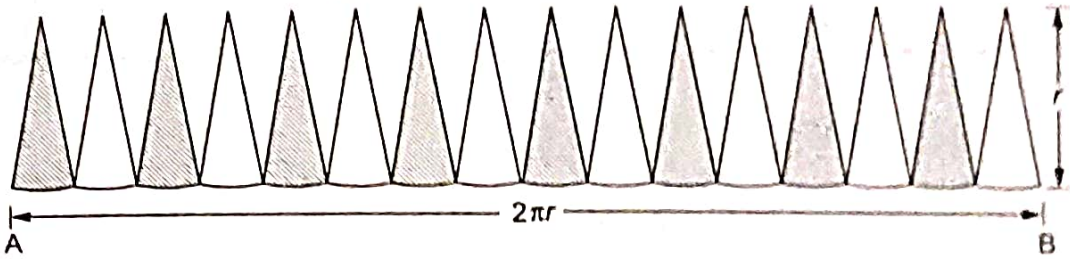

Fig. 33.2

Step 4. Now, paste the parts on the white paper to form a geometric figure closely resembling a parallelogram by arranging them in such a way that the shaded and unshaded parts appear alternately—the shaded parts pointing downwards while the unshaded parts pointing upwards as shown in Figure 33.3.

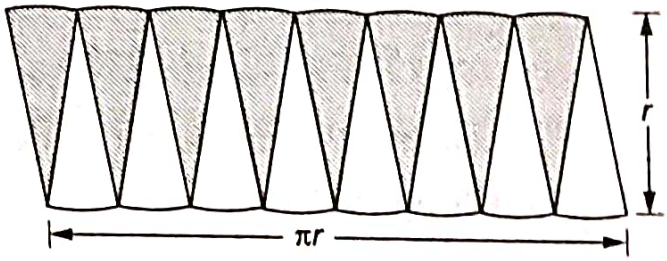

Fig. 33.3

Observations

We observe that the approximate length of the base of the parallelogram formed (in Figure 33.3) is πr units (i.e., half of the circumference of the circle in Figure 33.1) and the approximate height of this parallelogram is r units (i.e., equal to the radius of the circle in Figure 33.1).

$\therefore$ area of the circle (Figure 33.1) = area of the parallelogram (Figure 33.3)

$$= \pi r \times r$$

$$= \pi r^2.$$

Result The area (A) of a circle of radius r is given by $A = \pi r^2$.